CS781 Capstone Project:
In-House Legal Intelligence System

FirstName <Middle Name if any> LastName
Roll Number
{userid}@iitk.ac.in
Indian Institute of Technology Kanpur (IIT Kanpur)

Abstract

Write a concise abstract (maximum 150 words) summarizing: (i) the problem (confidential, self-hosted legal NLP); (ii) the seven-phase pipeline you built; (iii) your Phase-1 foundation model (size, architecture family, training data mix); (iv) how that foundation was (or was not) transferred to later phases; (v) the most important experimental findings. Do not list every hyperparameter here. Do not paste leaderboard screenshots into the abstract.

Important Guidelines (remove this entire section before final submission)

1. Follow the sections given below. The numbered sections are compulsory. You may add subsections and a modest number of extra figures/tables *inside* these sections. Do not rename, drop, or reorder the top-level sections.

2. **Maximum length: 10 pages** excluding references and the LLM Usage Declaration. Not even a single character more on the counted pages. References and the LLM declaration may use additional pages. This is a strict limit. A report that exceeds the limit will receive zero for the report component.

3. Write in continuous academic prose, not as a bullet dump. The itemized prompts under each section are a checklist of what the TAs expect to see; they are not a license to submit an outline.

4. There should be no grammar or spelling mistakes. Use a grammar checker.

5. Plagiarism is treated as an academic-integrity violation. Cite every paper, blog, repository, tutorial, and external snippet you used. Explain methods in your own words. Copied or closely paraphrased reports, and reports that are substantially similar across students, can lead to an F grade.

6. Cite papers with `\cite{...}` as in [1].

7. All mathematical expressions must be written in LaTeX.

8. Figures must be high resolution (vector PDF/SVG preferred). Do not screenshot tables from papers or from Codabench and paste them as images if you can typeset them. Every borrowed figure needs a citation.

9. Use LaTeX quotation marks: “like this”, not ”like this”.

10. In the final PDF, delete this guidelines section.

11. **LLM usage is allowed only in the limited ways stated in the capstone document.** Designing the model architecture, performing hyperparameter search, or writing the analysis/conclusions via an LLM is not allowed. Completing the LLM Usage Declaration is mandatory. Reports with a missing, blank, or inconsistent declaration will not be evaluated and will receive zero for the entire capstone.

12. Creativity, controlled ablations, honest negative results, and clear visualizations will be rewarded. A weaker model with rigorous analysis is worth more than an unexplained high score.

1 Competition Result

Codabench username: _____ **Roll number:** _____ Report the *best* *valid* score you obtained for each phase, together with the corresponding leaderboard rank at the time of report freeze.

Table 1: Official phase scores and ranks. Fill every cell. State the date on which ranks were recorded.

Phase	Task	Primary metric (value)	Rank	Valid submissions
1	Legal LM Pretraining			
2	Legal NER			
3	Rhetorical role prediction			
4	Court judgment prediction			
5	Legal statute identification			
6	Abstractive summarization			
7	Prior Case Retrieval (PCR)			

Total valid Codabench submissions across all phases: ____ **Date ranks were recorded:** ____ A minimum of 4 *distinct* submissions per phase is mandatory. Repeated identical dumps do not count. In one or two sentences, state whether you met the per-phase minimum and, if not, why.

2 Introduction and Problem Description

Cover the following in connected paragraphs (not as a numbered list in the submitted report):

- Motivation
- A precise statement of each task (input, output schema, primary metric).
- Constraints that shaped design: 50M–300M Phase-1 parameter cap, no pretrained LMs for the foundation, no external legal corpora, Colab/Kaggle-style compute, strict span matching for NER, multi-label statutes, BERTScore for summarization.

- High-level research questions you actually investigated (tokenizer, data mix, transfer from Phase 1, long-document encoding, decoding, thresholding, etc.).
- Scope statement: what you built end-to-end, and what you deliberately did not attempt.

3 Data Analysis and Preprocessing

This section is a grading criterion. Do not skip it or reduce it to "we loaded the Hugging Face dataset".

3.1 Sources and official splits

3.2 Corpus statistics that actually informed design

- Phase 1: token counts per source (Acts, High Courts, Supreme Court, Tribunals), document-length histograms, vocabulary / domain shift across sources, why you mixed sources the way you did (or why you trained on a subset).
- Phase 2: label frequency of the 12 entity types, span-length distribution, overlapping / nested / adjacent spans if present, boilerplate that produces false positives.
- Phase 3: role distribution in CL vs. IT, sentence length, and rare roles such as Dissent or PrecedentOverruled.
- Phase 4: class balance of 0 vs. 1, document length, whether outcome-bearing text sits at the end of the judgment.
- Phase 5: number of distinct statute IDs, labels per document (histogram), long-tail / rare statutes.
- Phase 6: document vs. summary length, compression ratio, n-gram overlap with the source (extractiveness).
- Phase 7: number of query documents, candidate pool size, average number of relevant prior cases per query, length distribution of queries vs. candidates, and any temporal or court-level structure in the official splits.

Include at least two figures or tables of your own (not screenshots of the dataset card).

3.3 Preprocessing

- Tokenization, lowercasing, punctuation, Unicode / Indian legal citations, whitespace that would shift Phase 2 character offsets.
- How you preserved character offsets for L-NER. This is a common source of silent zeros on the leaderboard; describe the alignment procedure.
- Sentence segmentation for Phase 3 and any mismatch with the provided sentence lists.
- Truncation / chunking / hierarchical packing for long documents in Phases 4–6.
- How you encoded query and candidate documents for Phase 7 (shared truncation policy, special tokens, or separate packing).

4 Approach and Model Evolution

Tell a chronological design story, then specify the final system. TAs should be able to re-implement the final pipeline from this section alone.

4.1 Phase-1 foundation model

- Architecture you *designed* (layers, heads, hidden size, FFN size, positional scheme, normalization, activation). Include a parameter count and a diagram. State how you verified the 50M–300M bound.
- Tokenizer: official vs. custom; vocab size; how legal citations / section symbols / party names are handled.
- Training objective, context / block length, data-mixing policy across the four sources, token budget actually consumed.
- Decoding used at submission time (greedy, temperature, top-k, top-p, repetition penalty, length targeting for 128 words).
- What you tried first, what failed.

4.2 Transfer protocol (Phases 2–7)

- Did you initialize later encoders from the Phase-1 checkpoint, train task models from scratch, or use a hybrid (frozen embeddings + task head, adapters, etc.)?
- Describe all finetuning technique you implemented in each phase. Discuss the results.
- A diagram of the overall system: one backbone, seven heads / seven separately fine-tuned descendants.

4.3 Phase 2 - Legal NER

- Span-extraction vs. token-classification-then-reconstruct-offsets. How character offsets are recovered exactly.
- Label set of 12 types; how you handled multi-span documents and class imbalance.
- Decoding: thresholding, n-best spans, overlap resolution.

4.4 Phase 3 - Rhetorical role prediction

- Sentence encoder and how much left / right context you condition on.
- Joint vs. per-sentence classification; document-level CRF / transition constraints if used.
- Domain shift between Corporate Law and IT, and whether you trained jointly or separately.

4.5 Phase 4 - Court judgment prediction

- Long-document encoder (truncation, CLS over chunks, hierarchical attention, use of Phase-2/3 structure as features).

4.6 Phase 5 - Legal statute identification

- Multi-label head (independent sigmoids, label-wise attention, etc.).
- Thresholding / top-k / tuned per-label thresholds.

4.7 Phase 6 - Abstractive summarization

- Encoder–decoder vs. decoder-only continuation; initialization from Phase 1.
- Long-document chunk-then-merge vs. hierarchical encode.
- Length control, coverage / repetition, and any post-training / alignment method (allowed if documented: data/feedback signal, how it was built from official data only, what changed).

4.8 Phase 7 - Prior Case Retrieval (PCR)

- Retrieval architecture (bi-encoder, cross-encoder, sparse+dense hybrid, or other).
- How query and candidate documents are encoded (shared encoder, separate towers, chunk aggregation).
- Handling of long documents.
- Failed approaches.

5 Experiments and Implementation Details

5.1 Compute and training protocol

- Hardware (Colab/Kaggle GPU type, any local GPU), wall-clock per run, number of runs, checkpointing strategy (runs will be killed).
- Optimizer, learning-rate schedule, warmup, batch / accumulation, dropout, weight decay, mixed precision, gradient clipping.
- Early stopping rule and which local metric it tracked (val loss, perplexity, span-F1, ...).

Table 2: Hyperparameters of the final model in each phase. Add rows as needed. One table per phase is acceptable if this becomes unreadable.

Hyperparameter	P1	P2	P3	P4	P5	P6	P7
Parameters (M)							
Context length							
Peak LR							
Batch × accum.							
Epochs / steps							
Dropout							
Decoding / threshold							

5.2 Local evaluation

- How you built a labelled held-out split from public train/dev data.
- Why local scores and Codabench scores diverged, if they did.

5.3 Ablations and controlled comparisons

You are expected to show *evidence* for each important claim ("custom tokenizer helped", "Phase-1 transfer helped", "hierarchical encoding helped"), give a comparison that changes one factor at a time. Negative ablations belong here; they are not a confession, they are the point of the course.

6 Results

- Local-dev tables for every major variant, with the primary metric and the official secondary metrics.
- Official Codabench numbers from Table 1.
- Learning curves: Phase-1 train/val loss or perplexity vs. tokens; task-phase metric vs. epoch.
- Which system won overall, and a causal explanation (not "it trained longer").

Table 3: Template for per-phase variant comparison

Phase	Variant	Codabench primary metric
1		
1		
2		
2		
...	...	...

7 Error Analysis and Discussion

Do not write "the model sometimes fails". Show examples, counts, and a hypothesized mechanism.

7.1 Phase-wise failure modes

7.2 What the metrics do not measure

A short, honest paragraph: ROUGE-L and BERTScore reward n-gram / embedding similarity to a hidden reference, not legal correctness. Strict span-F1 rewards offset perfection, not entity typing quality when the boundary is slightly off. Discuss the implications for a real firm deployment.

7.3 Limitations and what you would do next

Compute, context length, absence of external corpora, lack of human legal evaluation, error propagation if phases are pipelined rather than trained separately. Concrete next experiments, not vague "try transformers".

7.4 Surprising findings

Anything that contradicted your prior. One good unexpected result with evidence is worth more than three expected ones without it.

8 Conclusion

- The two or three design choices that moved metrics, and the ones that did not.
- The main lesson about training a compact legal LM under a parameter and data-source lock.

LLM Usage Declaration

Completing this section is mandatory. Inconsistent answers relative to the code and the prose will be treated as an integrity issue. **Q1.** Which Large Language Model(s), if any, did you use while completing the competition or writing this report? Give model name and vendor (e.g., none / ChatGPT-4.x / Gemini x / Claude x / Grok / Copilot / other).

Answer: _____ **Q2.**

For each row, mark Yes or No and state the purpose in one short phrase. "Yes" is acceptable only for the uses permitted by the capstone document.

Part	Used LLM?	Purpose of using the LLM
Understanding concepts / papers	Yes/No	
Ideation / brainstorming at a high level	Yes/No	
Debugging small ancillary snippets	Yes/No	
Plotting / visualization code	Yes/No	
Grammar / writing style of the report	Yes/No	
Report structure / section wording	Yes/No	
Designing the model architecture	Yes/No	
Hyperparameter search / tuning	Yes/No	
Analysis of results / conclusions	Yes/No	
Writing the core training / inference code	Yes/No	
Other (specify)	Yes/No	

Q3. Did you refer to sources other than lecture slides and an LLM? If yes, list them (papers, blogs, GitHub repositories, textbooks, official starter notebook, evaluation scripts). These must

also appear in the bibliography when they influenced the method.

Answer: **Q4.** Apart from any permitted LLM use, what was *your* contribution? In 8–12 sentences describe your own understanding, architecture decisions, implementation, experiments, and analysis. Name at least two things you tried that failed and what you concluded from them.

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in Neural Information Processing Systems*, vol. 30, 2017.